

The Kelly-Ulam graph reconstruction program aims to recover a graph from its deck of one-vertex-deleted subgraphs [2]. A standard bipartite version was formulated by Bondy and Hemminger as the problem of showing that bipartite graphs are reconstructible [1]. The following theorem proves a structured incidence-deletion analogue under strong type-distinguishability assumptions.

Terminology Let $\Omega = L \sqcup R$ be a finite vertex set with a partition into the disjoint parts $L, R \subseteq \Omega$. A graph K is bipartite with respect to (L, R) if every edge of K has one endpoint in L and one endpoint in R . We say G is 2-connected if G is connected and, for every vertex v , the induced graph on $\Omega \setminus \{v\}$ is connected. For a graph K and a vertex v , write $\Delta_v(K)$ for the graph obtained from K by deleting all edges incident to v , while leaving the vertex set unchanged. Thus

$$E(\Delta_v(K)) = \{xy \in E(K) : x \neq v, y \neq v\}.$$

A bipartite isomorphism $K \cong_B K'$ is a graph isomorphism whose underlying bijection preserves both parts L and R . The bipartite deck is the multiset

$$\mathcal{D}_B(K) := \{[\Delta_v(K)]_B : v \in \Omega\},$$

where $[\cdot]_B$ denotes bipartite-isomorphism class.

Let $N_K(x)$ be the neighbor set of x in K . The degree profile and type of x are

$$P_K(x) := \{\{\deg_K(y) : y \in N_K(x)\}\}, \quad \tau_K(x) := (\deg_K(x), P_K(x)).$$

The type profile of u is the multiset of the types of its neighbors:

$$T_K(u) := \{\{\tau_K(x) : x \in N_K(u)\}\}.$$

Finally, for a type $t = (d, M) \in \mathbb{N} \times \text{Multiset}(\mathbb{N})$ and $a \in \mathbb{N}$, define

$$F_a(t) := (d \dot{-} 1, M \setminus \{a\}),$$

where $d \dot{-} 1 = \max(d-1, 0)$ is truncated subtraction and $M \setminus \{a\}$ means deletion of one occurrence of a from the multiset M . In the applications below, F_a is applied to neighbor-types, whose first coordinate is positive, so $d \dot{-} 1 = d-1$ there. We write $\#_t M$ for the multiplicity of t in a multiset M .

Theorem (Weak bipartite graph reconstruction) Let G and H be finite simple graphs on the same finite vertex set Ω , both bipartite with respect to the same parts $\Omega = L \sqcup R$ and $|\Omega| \geq 3$. Assume that G is 2-connected and all vertex types of G are pairwise distinct:

$$x \neq y \implies \tau_G(x) \neq \tau_G(y).$$

If

$$\mathcal{D}_B(G) = \mathcal{D}_B(H),$$

then G and H are bipartite-isomorphic.

Proof. Write $n = |\Omega|$ and $e(K) = |E(K)|$. Since bipartite isomorphisms preserve edge counts, the deck determines the multiset

$$\{e(\Delta_v(K)) : v \in \Omega\}.$$

For every vertex v ,

$$e(\Delta_v(K)) + \deg_K(v) = e(K), \quad (1)$$

because $\Delta_v(K)$ removes exactly the edges incident to v . Also,

$$\sum_{v \in \Omega} e(\Delta_v(K)) = (n-2)e(K), \quad (2)$$

because each edge of K remains in precisely the cards indexed by the $n-2$ vertices not incident to that edge. Since $n \geq 3$, equation (2) shows that the deck determines $e(K)$. Applying this to the equal decks of G and H gives

$$e(G) = e(H), \quad (3)$$

and then equation (1) shows that the degree multisets of G and H are equal.

Equality of the two decks also gives a bijection $f : \Omega \rightarrow \Omega$ such that, for every $v \in \Omega$, there is a bipartite isomorphism

$$\phi_v : \Delta_v(G) \cong_B \Delta_{f(v)}(H). \quad (4)$$

For this matched pair of cards, edge counts are equal. Combining this with (1) and (3) yields

$$\deg_G(v) = \deg_H(f(v)) \quad (5)$$

for every v .

Next, G has minimum degree at least 2. Indeed, if a vertex had degree 0, then G would not be connected; if a vertex x had the unique neighbor y , then the induced graph on $\Omega \setminus \{y\}$ would contain the isolated vertex x , contradicting the assumed vertex-deletion connectivity. Since the degree multisets of G and H are equal, H also has minimum degree at least 2.

Therefore, in any graph K with minimum degree at least 2, the card $\Delta_v(K)$ has a unique isolated vertex, namely v . The vertex v is isolated because all its incident edges have been removed. If $x \neq v$, then x has at least two neighbors in K , at most one of which is v ; hence x has a neighbor $y \neq v$, and the edge xy remains in $\Delta_v(K)$.

Applying this uniqueness to (4), the isomorphism ϕ_v sends the unique isolated vertex of $\Delta_v(G)$ to the unique isolated vertex of $\Delta_{f(v)}(H)$. Hence

$$\phi_v(v) = f(v). \quad (6)$$

Since ϕ_v preserves L and R , (6) implies that f preserves the two parts:

$$v \in L \iff f(v) \in L, \quad v \in R \iff f(v) \in R. \quad (7)$$

We now prove that f preserves vertex types. It remains, in view of (5), to compare degree profiles. For a graph K and vertex v , put

$$A_K(k) := \#\{x \in \Omega : \deg_K(x) = k\},$$

$$B_{K,v}(k) := \#\{x \in \Omega : \deg_{\Delta_v(K)}(x) = k\},$$

and

$$C_{K,v}(k) := \#\{x \in N_K(v) : \deg_K(x) = k\}.$$

Counting vertices of degree k before and after deleting the incidence set of v gives, for every $k \geq 0$,

$$C_{K,v}(k+1) + A_K(k) + \mathbb{K}_{k=0} = C_{K,v}(k) + B_{K,v}(k) + \mathbb{K}_{\deg_K(v)=k}. \quad (8)$$

Since the global degree multisets are equal, $A_G(k) = A_H(k)$. Since ϕ_v is an isomorphism between the matched cards, $B_{G,v}(k) = B_{H,f(v)}(k)$. By (5), the final indicator in (8) is also the same for G and H . Finally, $C_{K,v}(0) = 0$, because a neighbor always has positive degree. Induction on k in (8) gives

$$C_{G,v}(k) = C_{H,f(v)}(k) \quad (k \geq 0).$$

Thus

$$P_G(v) = P_H(f(v)),$$

and together with (5) this proves

$$\tau_G(v) = \tau_H(f(v)) \quad (9)$$

for every $v \in \Omega$.

We next prove equality of type profiles:

$$T_G(u) = T_H(f(u)) \quad (10)$$

for every u . Fix u . If $u \in L$, take $S = R$; otherwise, since $\Omega = L \cup R$ and the parts are disjoint, take $S = L$. Thus S is the opposite side from u . By (7) and (9), the multisets of global types on S agree:

$$\{\{\tau_G(x) : x \in S\}\} = \{\{\tau_H(x) : x \in S\}\}. \quad (11)$$

The card isomorphism ϕ_u also preserves S , and therefore the multisets of local types on S agree:

$$\{\{\tau_{\Delta_u(G)}(x) : x \in S\}\} = \{\{\tau_{\Delta_{f(u)}(H)}(x) : x \in S\}\}. \quad (12)$$

For $x \in S$, bipartiteness gives the local-type formula

$$\tau_{\Delta_u(G)}(x) = \begin{cases} F_{\deg_G(u)}(\tau_G(x)), & \text{if } x \in N_G(u), \\ \tau_G(x), & \text{if } x \notin N_G(u). \end{cases} \quad (13)$$

Indeed, if x is adjacent to u , then deleting the incidence set of u removes the edge ux , lowering $\deg_G(x)$ by one and deleting one occurrence of $\deg_G(u)$ from the degree profile of x . If x is not adjacent to u , then any neighbor of x lies on the same side of the bipartition as u , and therefore cannot be adjacent to u ; hence the type of x is unchanged. The same formula holds for H and $f(u)$.

Taking the number of occurrences of an arbitrary type t in (13) gives

$$\#_t\{\{\tau_{\Delta_u(G)}(x) : x \in S\}\} + \#_t T_G(u) = \#_t\{\{\tau_G(x) : x \in S\}\} + \#_t\{\{F_{\deg_G(u)}(s) : s \in T_G(u)\}\}. \quad (14)$$

There is an identical identity for H . The first terms on the two sides of (14) agree by (12), and the global-type terms agree by (11). Also $\deg_G(u) = \deg_H(f(u))$ by (5). Hence equality of the type-profile counts can be proved by descending induction on the first coordinate of t . The base case is that no type in a type profile can have first coordinate at least n , since every vertex degree is $< n$. For the induction step, the only types s with $F_{\deg_G(u)}(s) = t$ have first coordinate $t_1 + 1$; moreover every s occurring in a type profile has positive first coordinate. Thus the rightmost term in (14) is already equal for G and H by the induction hypothesis, and subtracting the two versions of (14) yields equality of the number of occurrences of t in $T_G(u)$ and in $T_H(f(u))$. This proves (10).

The distinctness of types transfers from G to H : if $\tau_H(x) = \tau_H(y)$, then by applying (9) to $f^{-1}(x)$ and $f^{-1}(y)$ we obtain equality of the corresponding G -types, so $x = y$.

Since types are pairwise distinct, adjacency is determined by type profiles: for every graph K among G, H and every pair of vertices a, b ,

$$K \text{ has edge } ab \iff \tau_K(b) \in T_K(a). \quad (15)$$

The forward implication is the definition of $T_K(a)$. Conversely, if $\tau_K(b)$ occurs in $T_K(a)$, then some neighbor c of a has $\tau_K(c) = \tau_K(b)$, and distinctness of types gives $c = b$.

Finally, for any $u, v \in \Omega$, equations (9), (10), and (15) give

$$G \text{ has edge } uv \iff \tau_G(v) \in T_G(u) \iff \tau_H(f(v)) \in T_H(f(u)) \iff H \text{ has edge } f(u)f(v).$$

Thus f is a graph isomorphism $G \cong H$. By (7), it preserves the two bipartition parts, so it is a bipartite isomorphism.

References

- [1] J. A. Bondy and R. L. Hemminger. Graph reconstruction—a survey. *Journal of Graph Theory*, 1(3):227–268, 1977.
- [2] Paul J. Kelly. A congruence theorem for trees. *Pacific Journal of Mathematics*, 7(1):961–968, 1957.